

11.17 Not sure about the purpose of this problem. Do we need to derive the expression or just simply check the validity of it? Anyway, I will just check.

(a) The 4-velocity and the relative coordinates are

$$X^\alpha = (0, \vec{b} - \vec{v}t), \quad U^\alpha = (\gamma c, \gamma \vec{v})$$

Then, $X_\alpha U^\alpha = -\gamma \vec{v} \cdot (\vec{b} - \vec{v}t) = \gamma v^2 t$,

$$X_\alpha X^\alpha = -|\vec{b} - \vec{v}t|^2 = -(b^2 + v^2 t^2), \quad \text{since } \vec{b} \cdot \vec{v} = 0.$$

and $\frac{1}{c^2} (U_\alpha X^\alpha)^2 - X_\alpha X^\alpha = \frac{1}{c^2} \gamma^2 v^4 t^2 + b^2 + v^2 t^2 = b^2 + v^2 t^2 (1 + \gamma^2 \beta^2) = b^2 + \gamma^2 v^2 t^2.$

Let $\alpha = 0$, we know $F^{0i} = -E^i$. On the other hand,

$$F^{0i} = \frac{q}{c} \frac{X^0 U^i - X^i U^0}{\left[\frac{1}{c^2} (U_\alpha X^\alpha)^2 - X_\alpha X^\alpha \right]^{3/2}} = \frac{q}{c} \frac{-X^i U^0}{(b^2 + \gamma^2 v^2 t^2)^{3/2}}.$$

But $X^i U^0 = (\vec{b} - \vec{v}t)^i \cdot \gamma c = (b^i - v^i t) \gamma c$, then

$$E^i = -F^{0i} = \frac{q}{c} \frac{X^i U^0}{(b^2 + \gamma^2 v^2 t^2)^{3/2}} = \frac{q \gamma (b^i - v^i t)}{(b^2 + \gamma^2 v^2 t^2)^{3/2}},$$

which gives the first two relations in Eq. (11.152), if we notice that $\vec{b}$ and $\vec{v}$ can be chosen to align with $\hat{x}$ and $\hat{x}'$, respectively.

Let $\alpha, \beta = i, j$, we now have

$$\begin{aligned} F^{ij} &= \frac{q}{c} \frac{X^i U^j - X^j U^i}{\left[\frac{1}{c^2} (U_\alpha X^\alpha)^2 - X_\alpha X^\alpha \right]^{3/2}} = \frac{q}{c} \frac{(b^i - v^i t) \gamma v^j - (b^j - v^j t) \gamma v^i}{(b^2 + \gamma^2 v^2 t^2)^{3/2}} \\ &= \frac{q}{c} \frac{\gamma (b^i v^j - b^j v^i)}{(b^2 + \gamma^2 v^2 t^2)^{3/2}} = \frac{q}{c} \frac{\gamma \hat{e}^k \epsilon^{ijk} b^i v^j}{(b^2 + \gamma^2 v^2 t^2)^{3/2}} = -\frac{q}{c} \frac{\gamma (\vec{v} \times \vec{b})^k}{(b^2 + \gamma^2 v^2 t^2)^{3/2}} \end{aligned}$$

But $F^{ij} = -\epsilon^{ijk} B^k$, and $\vec{B} = \frac{q \gamma (\vec{\beta} \times \vec{b})}{(b^2 + \gamma^2 v^2 t^2)^{3/2}}$, which agrees with the third equation of (11.152)

(b) In the reference frame K' , we have $\gamma'^\alpha = (0, \vec{b} - \vec{v}t')$, $U^\alpha = (c, 0)$. Then,

$$\gamma_2 \gamma^\alpha = -|\vec{b} - \vec{v}t'|^2 = -(b^2 + v^2 t'^2) = -r'^2, \text{ and}$$

$$F^{0i} = \frac{q}{c} \frac{\gamma'^0 U^i - \gamma'^i U^0}{(-\gamma_2 \gamma_2)^{3/2}} = -\frac{q}{c} \frac{-(b^i - v^i t')c}{r'^3} = \frac{q(b^i - v^i t')}{r'^3},$$

$$F^{ij} = \frac{q}{c} \frac{\gamma'^i U^j - \gamma'^j U^i}{(-\gamma_2 \gamma_2)^{3/2}} = 0,$$

which give the field strength in the reference frame K' .

(c) Following the same argument as in Section 14.1, consider the configuration on the right,

We must have $(vt - \beta R)^2 + b^2 = R^2$, or $\beta^2 (ct - R)^2 + b^2 = R^2$.

This is equivalent to $(1 - \beta^2)R^2 + 2\beta^2 ctR - (b^2 + \beta^2 c^2 t^2) = 0$,

$$\text{or } R^2 + 2\gamma^2 \beta^2 ctR - \gamma^2 (b^2 + \beta^2 c^2 t^2) = 0.$$

The physical solution to this equation is given by

$$R = -\gamma^2 \beta^2 ct + \sqrt{\gamma^4 \beta^4 c^2 t^2 + \gamma^2 b^2 + \gamma^2 \beta^2 c^2 t^2}.$$

Notice that $\gamma^4 \beta^4 c^2 t^2 + \gamma^2 \beta^2 c^2 t^2 = \gamma^2 \beta^2 c^2 t^2 (1 + \gamma^2 \beta^2) = \gamma^4 v^2 t^2$, then

$$R = \gamma [b^2 + \gamma^2 v^2 t^2]^{1/2} - \gamma^2 \beta^2 ct.$$

Now, $U^\alpha = (\gamma c, \gamma \vec{v})$, $Z^\alpha = (R, \vec{b} - \vec{\beta}(ct - R))$, we have

$$U_\alpha Z^\alpha = \gamma c R + \gamma \beta^2 c (ct - R) = \gamma \beta^2 c^2 t + \frac{1}{\gamma} c R = c [b^2 + \gamma^2 v^2 t^2]^{1/2}.$$

$$\text{Also, } Z^0 U^i - Z^i U^0 = R \gamma v^i - (b^i - \beta^i (ct - R)) \gamma c = -\gamma c (b^i - v^i t),$$

$$\text{and } Z^i U^j - Z^j U^i = (b^i - \beta^i (ct - R)) \gamma v^j - (b^j - \beta^j (ct - R)) \gamma v^i = \gamma (b^i v^j - b^j v^i),$$

which gives identical result as in part (a).

